

Shadow of HyperPose: New Animation System

New Animation System

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ABSTRACT

This presentation details HyperPose, an innovative animation system that represents motion through strategically chosen samples (“principal dynamic poses”) integrated into a procedural state machine. State navigation is achieved via a cost function, optimizing the system’s efficiency and realism in animation. Data is represented as a four-dimensional hyperpose.

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Circa 2023, the primary animation systems in video games are “Animation State Machine (ASM)” and “Motion Matching (MM).” While these approaches offer immense utility, we will discuss their inherent and unavoidable limitations concerning quality, responsiveness, and artistic control.

The core concept of HyperPose lies in increasing the dimensionality of data (Guo et al., 2021) (McGuirk, 2021) and saturating the result with N samples (Figure 1). Human posing capacity is infinite yet limited by anatomy and physics. This implies that a collection of all poses achievable in principle can be described with a finite number of samples, termed “principal dynamic poses (PDPs),” and their interpolated derivatives.

The idea of segmenting an animation sequence to obtain poses that can be further used to generate a new blend tree has been explored multiple times in the past (Kovar et al., 2002) (Buttner, 2017) (Lee et al., 2010). Our method of motion sampling suggests that such a “complete” hyperpose can be described in full using about 1500 PDPs, replacing the need to store extensive animation data with a compact asset of extra dimension and respective metadata, not dissimilar from a blend space.

After identifying the PDPs, we conduct a dynamic pose comparison, calculating similarity costs for each PDP relative to all others. Consequently, any PDP can be substituted with a pointer to its most similar counterpart, enabling footprint scaling based on the desired size. This allows for multiple branching paths at each node of the chain, each with pre-calculated costs. This results in a procedural state machine whose nodal connectivity is one to two orders

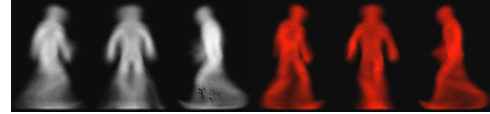


Figure 1: Locomotion hyperpose and a single sample of it.

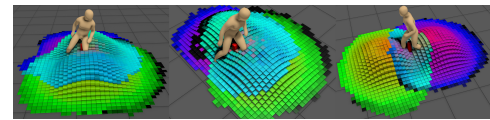


Figure 2: Heatmap visualization of space of possibilities in the past (blue) and future (green), available within 1 second in either direction. The amount of red signifies proximity to the current state in time.

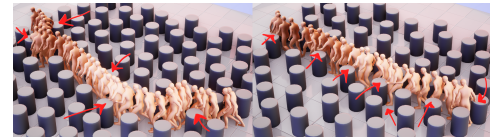


Figure 3: HyperPose solutions for character navigating a room full of obstacles, with full contact, as an optimal sequence of PDPs.

of magnitude greater than that found in current ASMs. However, with the limited number of PDPs and each being readily accessible, we maintain direct and explicit control over any derivative pose achieved, which is unavailable to this extent in MM (Figure 2).

We employ a multi-step cost function to find the optimal sequence of PDPs to execute based on any current and target states. The interpolation of PDPs yields motion reminiscent of the progression seen in traditional keyframe animation (we are considering a bidirectional inertialization). PDP attributes encompass spatial and temporal offsets, collider transformations, custom tags, and any additional relevant information. Consequently, all these properties can be integrated into the decision-making process: HyperPose solutions include classic use cases but also unconventional ones, such as audio to full body motion, dense collision navigation, paired locomotion, and others (Figure 3).

We conclude the talk with a review of the current stage of development, known advantages and concerns, and future steps, showcasing details and examples that are impossible to include in this brief abstract.

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